Monte Carlo Pricing of a European Call: Euler–Maruyama Discretisation under Geometric Brownian Motion

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Abstract

We specialise the Euler–Maruyama scheme of the previous theoretical writeup to geometric Brownian motion, derive the resulting recursion, and apply it to Monte Carlo pricing of the European call. The algorithm is biased: unlike the exact GBM sampler, the estimator converges to $\mathbb{E}[\Phi(\hat{S}_T)]$, not to $\mathbb{E}[\Phi(S_T)]$, and the gap is the discretisation bias of order h predicted by the weak convergence theorem. The principal contribution of this writeup is the design of the empirical convergence study that validates the predicted strong order $\frac{1}{2}$ and weak order 1 on a controlled benchmark, anchored on the closed-form GBM solution that allows pathwise comparison between the discretised and the exact dynamics.

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1 Introduction

The previous writeup (`sde_discretisation.tex`) developed the abstract theory of strong and weak convergence for SDE discretisation schemes, derived the Euler–Maruyama scheme as a first-order truncation of the Itô-Taylor expansion, and stated its convergence orders: strong $\frac{1}{2}$ and weak 1. The present document specialises that theory to the simplest non-trivial setting in finance: pricing a European call under geometric Brownian motion, where the underlying SDE admits a closed-form solution and a direct comparison between discretised and exact paths is therefore possible.

The setting is degenerate in the sense that no one would actually use Euler to price a European call under GBM: the exact-simulation algorithm, which samples S_T exactly with one normal draw per path, is both faster (no inner time loop) and exact (no discretisation bias). The value of the present algorithm is twofold. First, it is the simplest laboratory in which to verify empirically that the convergence orders predicted by Theorem 5.1 of the previous writeup are real and quantitatively correct, on a problem where the exact solution is known and pathwise comparison is feasible. Second, the Euler scheme implemented here is the foundation on which all SDE-driven Monte Carlo work in the later notes rests: when local volatility, or jump-diffusion, or stochastic volatility models are added, the structure of the pipeline (sample increments, recurse the state, evaluate payoff at terminal value, average) is unchanged; only the recursion changes.

The remainder of this document is organised as follows. Section 2 restates the SDE setting and specialises Euler–Maruyama to GBM. Section 3 defines the biased Monte Carlo estimator and decomposes its mean square error into bias and variance contributions. Section 4 designs the empirical convergence study. Section 5 states the algorithm formally. Section 6 addresses the practical question of choosing h at fixed computational budget.

2 Euler–Maruyama for geometric Brownian motion

2.1 The recursion

Recall the GBM SDE under the risk-neutral measure $\mathbb{Q}$:

$$\mathrm{d}S_t \;=\; rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0. \tag{1}$$

With $a(s) = rs$, $b(s) = \sigma s$, the Euler–Maruyama recursion (equation (27) of the SDE discretisation note) specialises to

$$\hat{S}_{n+1} \;=\; \hat{S}_n \;+\; r\,\hat{S}_n\,h \;+\; \sigma\,\hat{S}_n\,\Delta W_n \;=\; \hat{S}_n\,\big(1 + rh + \sigma\,\Delta W_n\big), \qquad \hat{S}_0 = S_0, \tag{2}$$

with the Brownian increments $\Delta W_n = W_{(n+1)h} - W_{nh} \sim \mathcal{N}(0, h)$ independent across $n = 0, 1, \ldots, N-1$, and $h = T/N$.

The recursion is multiplicative: $\hat{S}_{n+1}$ is $\hat{S}_n$ multiplied by the random factor $(1 + rh + \sigma\Delta W_n)$. As a consequence,

$$\hat{S}_T \;=\; \hat{S}_N \;=\; S_0 \prod_{n=0}^{N-1} \big(1 + rh + \sigma\Delta W_n\big). \tag{3}$$

This is a discrete-time, geometric-multiplicative process.

2.2 Failure of positivity

The exact GBM solution is positive almost surely: $S_t = S_0 \exp((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t) > 0$. The Euler recursion does not preserve this property. Whenever $1 + rh + \sigma\Delta W_n < 0$, we have $\hat{S}_{n+1}$ of the *opposite sign* from $\hat{S}_n$. This requires

$$\Delta W_n \;<\; -\frac{1 + rh}{\sigma}, \tag{4}$$

an event that, since $\Delta W_n \sim \mathcal{N}(0, h)$, has probability

$$\Phi\left(-\frac{1+rh}{\sigma\sqrt{h}}\right). \tag{5}$$

For $r = 0.05$, $\sigma = 0.20$, and $h = 1/100$, this probability is approximately $\Phi(-50.25) \approx 10^{-553}$: astronomically small, operationally zero. For $h = 1/10$ and the same parameters, the threshold is $\Phi(-15.96) \approx 10^{-58}$: still operationally zero. For very large σ or very small N the threshold relaxes but remains far in the tail.

We will not implement any defence against this event in the algorithm itself: the empirical probability is small enough that no run we will perform is expected to encounter it. If the simulation grids are extended to include very coarse h or very high volatility, a positivity check can be added at little cost. A numerically prudent alternative is the *log-Euler* scheme, which applies Euler in $\log S$ and then exponentiates; for GBM this turns out to coincide with the exact GBM sampler, so it is not informative as a test of the Euler convergence rate.

2.3 Convergence orders

By Theorem 5.1 of the previous writeup, applied to the GBM coefficients which satisfy the regularity assumptions, the Euler recursion (2) converges to the exact S_T with

$$\mathbb{E}\big[\,|S_T - \hat{S}_T|\,\big] \;=\; O(h^{1/2}), \qquad |\mathbb{E}[f(S_T)] - \mathbb{E}[f(\hat{S}_T)]| \;=\; O(h) \quad \text{for } f \text{ smooth enough.} \tag{6}$$

The first is strong order $\frac{1}{2}$, the second is weak order 1. Both will be verified empirically in Section 4.

3 The biased Monte Carlo estimator

3.1 Definition

Let $\hat{S}_T^{(i)}$ for $i = 1, \ldots, M$ denote independent realisations of the Euler-discretised terminal price, each generated by the recursion (2) with its own sequence of Brownian increments. The Monte Carlo Euler estimator of the European call price is

$$\hat{C}_{M,h}^{\text{Euler}} \;\stackrel{\text{def}}{=}\; \frac{1}{M}\sum_{i=1}^{M} e^{-rT}\left(\hat{S}_T^{(i)} - K\right)^+. \tag{7}$$

Two sources of error contribute to $\hat{C}_{M,h}^{\text{Euler}} - C_{\text{BS}}$: a discretisation bias from replacing S_T with $\hat{S}_T$, and a sampling error from finite M.

3.2 Bias–variance decomposition

Define

$$C_{\text{Euler}}(h) \;\stackrel{\text{def}}{=}\; \mathbb{E}\big[\,e^{-rT}\,(\hat{S}_T - K)^+\,\big], \tag{8}$$

the population mean of the discretised payoff at step size h. The estimator decomposes as

$$\hat{C}_{M,h}^{\text{Euler}} - C_{\text{BS}} \;=\; \big(\hat{C}_{M,h}^{\text{Euler}} - C_{\text{Euler}}(h)\big) \;+\; \big(C_{\text{Euler}}(h) - C_{\text{BS}}\big). \tag{9}$$

The first term has zero mean and standard deviation $\sqrt{\text{Var}(\hat{S}_T \cdot \text{payoff})/M}$, by the same central limit theorem argument as for the exact GBM pricer: it is the statistical noise controlled by sample size M. The second term is deterministic and does not vanish as $M \to \infty$: it is the discretisation bias controlled by step size h, and by (6) it is $O(h)$.

The mean square error of the estimator is therefore

$$\mathbb{E}\big[\,(\hat{C}^{\text{Euler}}_{M,h} - C_{\text{BS}})^2\,\big] \;=\; \frac{\sigma^2_{\text{Euler}}(h)}{M} \;+\; \big(C_{\text{Euler}}(h) - C_{\text{BS}}\big)^2 \;\sim\; \frac{C_1}{M} + C_2\,h^2, \tag{10}$$

where $\sigma^2_{\text{Euler}}(h)$ is the variance of the Euler-discretised discounted payoff (which converges to the exact $\text{Var}(Y)$ of the exact GBM pricer as $h \to 0$, hence the constant approximation C_1), and the second term reflects the squared bias.

This decomposition is the conceptual novelty of SDE discretisation with respect to exact simulation, where the bias is identically zero. Choosing the simulation parameters now requires balancing two effects rather than one. We address the practical implications in Section 6.

4 Empirical validation strategy

This section is the heart of the document. We design four tests that together verify both the theoretical predictions of Theorem 5.1 of the previous writeup and the implementation correctness. Two of the tests measure convergence orders directly (strong and weak); the third anchors the implementation in the existing exact pricer; the fourth checks a basic invariance.

4.1 Test 1: empirical strong order

The strong order is the rate of decay of $\mathbb{E}[|S_T - \hat{S}_T|]$ in h. To estimate this expectation, we need to evaluate S_T and $\hat{S}_T$ *on the same Brownian path*. The trick that makes this feasible is the closed-form GBM solution: given the simulated Brownian increments $\Delta W_0, \ldots, \Delta W_{N-1}$ used by the Euler scheme, the corresponding exact terminal value is

$$S_T^{\text{exact}} \;=\; S_0 \exp\!\big(\big(r - \tfrac{1}{2}\sigma^2\big)T + \sigma\, W_T\big)\,, \qquad W_T = \sum_{n=0}^{N-1} \Delta W_n. \tag{11}$$

This S_T^{exact} is the value of the true SDE solution at T for the very same realisation of Brownian motion that drove the Euler simulation. The pathwise difference $|S_T^{\text{exact}} - \hat{S}_T|$ is therefore a sample of the strong error.

The protocol is:

1. Choose a logarithmic grid of step counts $N_1 < N_2 < \ldots < N_K$, equivalently $h_k = T/N_k$.
2. For each N_k, generate M independent Euler paths, each with its own sequence of N_k Brownian increments.
3. For each path i, compute $S_T^{\text{exact},(i)}$ from (11) and $\hat{S}_T^{(i)}$ from the Euler recursion.
4. Estimate $\text{strong}(h_k) \approx M^{-1} \sum_{i=1}^{M} |S_T^{\text{exact},(i)} - \hat{S}_T^{(i)}|$.
5. Plot $\log(\text{strong}(h_k))$ against $\log(h_k)$.

The expected behaviour is a straight line of slope close to $\frac{1}{2}$. Substantial deviation from this slope indicates either a bug in the Euler implementation, in the exact comparison, or a regime where the asymptotic order has not yet been reached (very small N).

4.2 Test 2: empirical weak order

The weak order is the rate of decay of $|\mathbb{E}[f(S_T)] - \mathbb{E}[f(\hat{S}_T)]|$ in h, for the call payoff $f(s) = e^{-rT}(s-K)^+$ in our setting.

1. Choose the same grid of step counts N_k.

2. For each N_k, generate M Euler paths and form $\hat{C}^{\text{Euler}}_{M,h_k}$ from (7).

3. Estimate $\text{weak}(h_k) \approx |\hat{C}^{\text{Euler}}_{M,h_k} - C_{\text{BS}}|$.

4. Plot $\log(\text{weak}(h_k))$ against $\log(h_k)$, fit a straight line.

The expected slope is close to 1. The two convergence studies together verify the asymmetry of the SDE convergence orders, the single most important conceptual takeaway of SDE discretisation.

4.3 Calibration of M for the convergence studies

Tests 1 and 2 are subtle: they estimate quantities that are themselves expectations, so the Monte Carlo noise $O(M^{-1/2})$ acts as a floor on what slopes can be measured.

For the strong test, the empirical strong error $\text{strong}(h)$ has standard error of order $\sqrt{\text{Var}(|S_T^{\text{exact}} - \hat{S}_T|)/M}$. As long as this standard error is small compared with the strong error itself, the slope is well-defined. For $h \in [10^{-4}, 10^{-1}]$ and our GBM parameters, $|S_T^{\text{exact}} - \hat{S}_T|$ has typical scale 10^{-1} to 10^{-2} and standard deviation comparable; setting $M = 10^4$ gives a relative standard error of 10^{-2}, sufficient to resolve the $\frac{1}{2}$ slope across two orders of magnitude in h.

For the weak test, the empirical weak error $\text{weak}(h)$ has standard error of order $\sigma_Y/\sqrt{M}$ where σ_Y is the standard deviation of the discounted payoff. For our GBM parameters $\sigma_Y \approx 14$ and the weak error is $O(h)$ which can be as small as 10^{-3}. To resolve the slope 1 we need $\sigma_Y/\sqrt{M} \ll 10^{-3}$, i.e. $M \gg 2 \times 10^8$. This is impractical. The standard remedy is *common random numbers between the simulated and the reference computations*: instead of comparing $\hat{C}^{\text{Euler}}_{M,h}$ with the closed-form C_{BS}, we compute both estimators (Euler and exact) on the same Brownian paths and look at their difference, which has much smaller variance than either estimator individually.

Specifically, with the same M Brownian paths, evaluate

$$\hat{D}_{M,h} \overset{\text{def}}{=} \frac{1}{M} \sum_{i=1}^{M} e^{-rT} \left[(\hat{S}_T^{(i)} - K)^+ - (S_T^{\text{exact},(i)} - K)^+ \right]. \tag{12}$$

Then $\mathbb{E}[\hat{D}_{M,h}] = C_{\text{Euler}}(h) - C_{\text{BS}}$, which is exactly the weak error, and $\text{Var}(\hat{D}_{M,h}) = M^{-1} \text{Var}(\Phi(\hat{S}_T) - \Phi(S_T))$, which is much smaller than $M^{-1} \text{Var}(\Phi(\hat{S}_T))$ because the two payoffs are highly correlated under common Brownian. With $M = 10^5$ this is sufficient to resolve the slope.

4.4 Test 3: coherence with the exact pricer

This test is operational rather than theoretical: it confirms that as $h \to 0$ at fixed M, the Euler pricer's estimate approaches the exact GBM pricer's estimate (modulo the irreducible $O(M^{-1/2})$ sampling noise). A failure here would signal a bug, not a violation of the theory.

The cleanest way to run this test under common random numbers is: generate one batch of M Brownian paths at N steps each, run the Euler scheme on them, and compare the resulting estimator against the exact estimator built from $W_T = \sum_n \Delta W_n$ on the same paths. The two estimators share noise; their difference is at most the discretisation bias.

4.5 Test 4: input validation and basic invariances

A separate, mechanical battery of tests checks that the implementation rejects bad inputs (negative S, K, σ, T, non-positive N or M), that the result is monotone in the parameters in the expected directions (modulo noise), and that the asymptotic limits $S_0 \to \infty$ and $S_0 \to 0$ are respected, mirroring Test 4 of the exact GBM pricer.

5 Algorithmic specification

The algorithm is summarised below. The pseudocode is parametric in the random source, following the seed-control discipline established in the Monte Carlo foundations.

Algorithm 1 Euler–Maruyama Monte Carlo for the European call (GBM)

Require: Spot S_0, strike K, maturity T, rate r, volatility σ, step count N, sample size M, confidence level $1-\beta$, random generator `rng` initialised with a recorded seed.
Ensure: Estimate $\hat{C}$, half-width h_M, sample variance S_M^2.

$h \leftarrow T/N, \quad s \leftarrow \sigma\sqrt{h}, \quad D \leftarrow e^{-rT}$
for $i = 1, \dots, M$ **do**
  $S^{(i)} \leftarrow S_0$
  for $n = 0, 1, \dots, N-1$ **do**
    Draw $\Delta W \sim \mathcal{N}(0, h)$ using `rng`
    $S^{(i)} \leftarrow S^{(i)}\,(1 + rh + \sigma\,\Delta W)$
  end for
  $Y_i \leftarrow D \cdot \max(S^{(i)} - K,\, 0)$
end for
$\hat{C} \leftarrow M^{-1} \sum_i Y_i$
$S_M^2 \leftarrow (M-1)^{-1} \sum_i (Y_i - \hat{C})^2$
$h_M \leftarrow z_{\beta/2} \sqrt{S_M^2/M}$
return $(\hat{C},\, h_M,\, S_M^2)$

The implementation will collapse the inner loop into a single vectorised pass over all M paths simultaneously, advancing the state $\hat{S}_n$ as a length-M array. The pseudocode is written elementwise for clarity only.

6 Practical considerations: choosing h and M

By the MSE decomposition (10),

$$\mathbb{E}\big[\,(\hat{C}_{M,h}^{\mathrm{Euler}} - C_{\mathrm{BS}})^2\,\big] \;\sim\; \frac{C_1}{M} + C_2\,h^2. \tag{13}$$

The total computational cost per path is proportional to $N = T/h$ (the inner time loop), so the overall cost of the simulation is proportional to $M \cdot N = MT/h$. To minimise MSE at a fixed budget $W = M/h$ (proportional to total cost), set $L(h) = C_1 h/W \cdot (\text{rearrangement}) + C_2 h^2$, take the derivative with respect to h, and equate to zero. The optimal allocation balances $\frac{C_1}{M} \sim C_2 h^2$, i.e. $M \sim h^{-2}$, equivalent to

$$M^*(h) \;\propto\; h^{-2}, \quad \text{or equivalently,} \quad h^*(M) \;\propto\; M^{-1/2}. \tag{14}$$

At the optimum, both terms in the MSE are of the same order, and the achieved error scales as $W^{-1/3}$ in computational budget W, markedly slower than the unbiased $W^{-1/2}$ rate of exact simulation.

This degradation is the price of using Euler. The fact is well-known and is, again, the reason no one would actually use Euler for European pricing under GBM: the unbiased exact GBM sampler delivers $W^{-1/2}$, an order of magnitude better at large budgets. The Euler scheme is operationally relevant when no closed-form solution exists (local volatility, stochastic volatility, jump-diffusion), in which case the comparison is not between Euler and exact but between Euler and other discretisation schemes.

This analysis is also the foundation for the multilevel Monte Carlo techniques of Giles (2008), which combine simulations at multiple step sizes in a way that recovers $W^{-1/2}$ even with biased samplers. Multilevel MC is beyond the scope of these notes but appears prominently in the path-dependent literature.

7 Bibliographic notes

The empirical convergence study reproduced in this writeup is the classical demonstration of Higham [2], with adapted parameters. Glasserman [1, Chapter 6] treats Euler specifically for finance applications and includes the bias–variance analysis of Section 6. Kloeden and Platen [3] provide the formal proofs of the convergence orders cited from the previous theoretical writeup.

References

[1] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

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